Taller 1, punto 10: Adaline sobre las particiones de Iris

A) setosa vs resto B) versicolor vs virginica. Salida en [0,1], umbral 0.5

```
clear; close all; clc
load iris_splits.mat
```

Parametros

```
umbral = 0.5; salida = [0 1]; error_min = 1e-3; max_epocas = 500; semilla = 1;
alphas = [0.001 0.005 0.01 0.05 0.1];
problemas = {'A: setosa vs resto', 'B: versicolor vs virginica'};
```

Entrenamiento

```
R = table();
for pr = 1:2
    for k = 1:4
        if pr == 1
            Xtr = S(k).Xtr; Xte = S(k).Xte;
            dtr = double(S(k).ytr == 1);  dte = double(S(k).yte == 1);
        else
            mtr = S(k).ytr ~= 1; mte = S(k).yte ~= 1;
            Xtr = S(k).Xtr(mtr,:); Xte = S(k).Xte(mte,:);
            dtr = double(S(k).ytr(mtr) == 2);  dte = double(S(k).yte(mte) == 2);
        end
        mn = min(Xtr); mx = max(Xtr);
        Xtr = (Xtr - mn) ./ (mx - mn);  Xte = (Xte - mn) ./ (mx - mn);
        for a = alphas
            [w, b, ep, EC] = adaline(Xtr, dtr, a, error_min, max_epocas, semilla);
            acc_tr = 100*mean(predecir(Xtr, w, b, umbral, salida) == dtr);
            acc_te = 100*mean(predecir(Xte, w, b, umbral, salida) == dte);
            R = [R; table(problemas(pr), k, {sprintf('%d-%d', 100*S(k).p,
100-100*S(k).p)}, a, ep, EC(end), acc_tr, acc_te, ...
                 'VariableNames',
{'Problema','k','Particion','Alpha','Epocas','EC_final','Exact_train','Exact_test'})
];
            if k == 1 && a == 0.01, curva{pr} = EC; end
        end
    end
end
R
```

R = 40×8 table

...

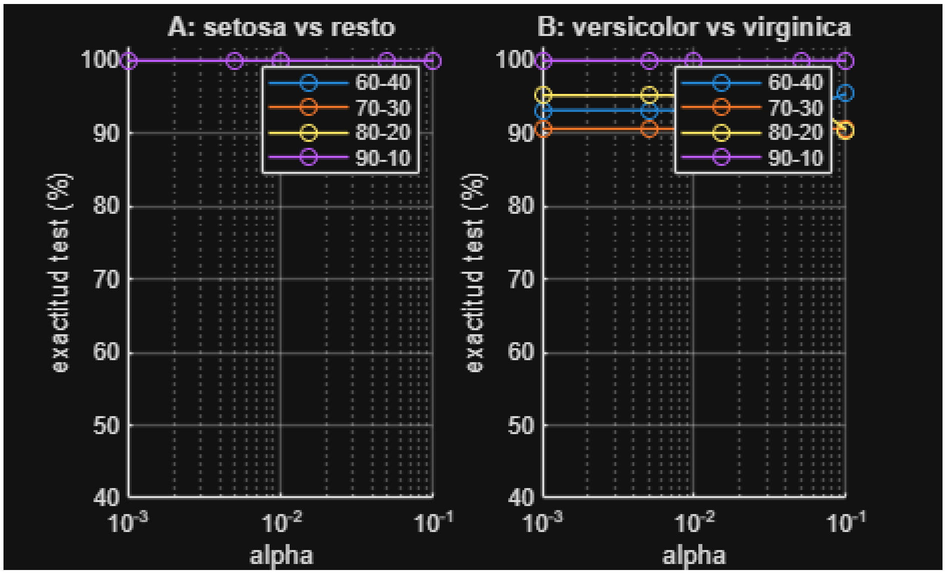
	Problema	k	Particion	Alpha	Epocas	EC_final
1	'A: setosa vs resto'	1	'60-40'	1.0000e-03	500	0.9592
2	'A: setosa vs resto'	1	'60-40'	0.0050	500	0.9453

	Problema	k	Particion	Alpha	Epocas	EC_final
3	'A: setosa vs resto'	1	'60-40'	0.0100	500	0.9485
4	'A: setosa vs resto'	1	'60-40'	0.0500	500	0.9662
5	'A: setosa vs resto'	1	'60-40'	0.1000	337	1.0079
6	'A: setosa vs resto'	2	'70-30'	1.0000e-03	500	1.1164
7	'A: setosa vs resto'	2	'70-30'	0.0050	500	1.1078
8	'A: setosa vs resto'	2	'70-30'	0.0100	500	1.1114
9	'A: setosa vs resto'	2	'70-30'	0.0500	464	1.1272
10	'A: setosa vs resto'	2	'70-30'	0.1000	290	1.1792
11	'A: setosa vs resto'	3	'80-20'	1.0000e-03	500	1.3227
12	'A: setosa vs resto'	3	'80-20'	0.0050	500	1.3045
13	'A: setosa vs resto'	3	'80-20'	0.0100	500	1.3028
14	'A: setosa vs resto'	3	'80-20'	0.0500	458	1.3071

⋮

Graficas

```
figure
for pr = 1:2
    subplot(1,2,pr); hold on
    for k = 1:4
        m = strcmp(R.Problema, problemas{pr}) & R.k == k;
        plot(R.Alpha(m), R.Exact_test(m), '-o', 'DisplayName',
R.Particion{find(m,1)});
    end
    set(gca, 'XScale', 'log'); grid on; ylim([40 102]); legend
    xlabel('alpha'); ylabel('exactitud test (%)'); title(problemas{pr});
end
```




```
figure
semilogy(curva{1}); hold on; semilogy(curva{2}); grid on
xlabel('epoca'); ylabel('EC'); legend(problemas); title('Particion 60-40, alpha =
0.01')
```